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Assignment - 3

SECTION - A

1) Clocking methodology is a synchronization mechanism in processors where all operations are coordinated using a clock signal. Each clock pulse triggers data movement b/w registers and execution in functional units, ensuring orderly and timed execution of instructions.

2) A clock cycle is the smallest unit of time in processor operation. Each instruction is broken into steps executed in one or more clock cycles. Processor speed is inversely proportional to clock cycle time.

3) Amdahl's law: defines the theoretical speedup of a sys. when a part of it is improved.

$$\text{speedup} = \frac{1}{(1-P) + (P/S)}$$

where, P = parallel portion of the program
S = speedup of that ~~operation~~ portion

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4) The datapath consists of hardware components responsible for processing data such as ALU, registers, buses & multiplexers. It is the part of the CPU that actually executes the instructions

5) The control unit directs the opeⁿ of the processor by generating control signals. It determines which opⁿ the ALU performs, when data is moved b/w components & which instruction is executed.

6) A ~~single~~ ~~#~~ cycle processor executes an entire instruction in one clock cycle. It has a simple design but requires a long clock cycle, making it slower & less efficient.

7) A multicycle processor breaks the instruction execution into multiple steps. It has a shorter clock cycle time and better hardware utilization compared to ~~signal~~ single cycle processor.

8) Instruction pipelining is a technique in which the execution of multiple instructions is overlapped by dividing the instruction cycle into a number of stages. This increases throughput and CPU performance.

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9) ILP is the ability of a processor to execute multiple instructions simultaneously by exploiting independent instructions in the program.

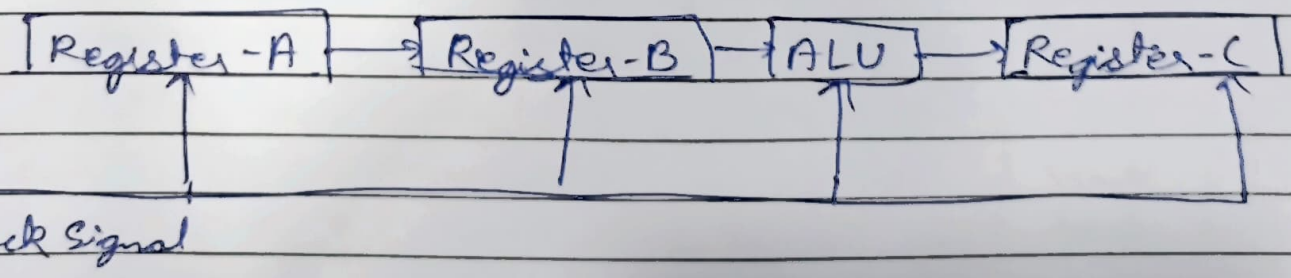
10) Data hazards occur in pipelining when an instruction depends on the result of a previous instruction which has not completed its execution.

11) Control hazards occur due to branch instructions which change the normal flow of program.

12) Pipeline hazards can be mitigated using
- Forwarding
- Pipeline stalls
- Branch prediction
- Instruction re-ordering.

SECTION - B

1)



It synchronizes opⁿ of all registers & functional units.

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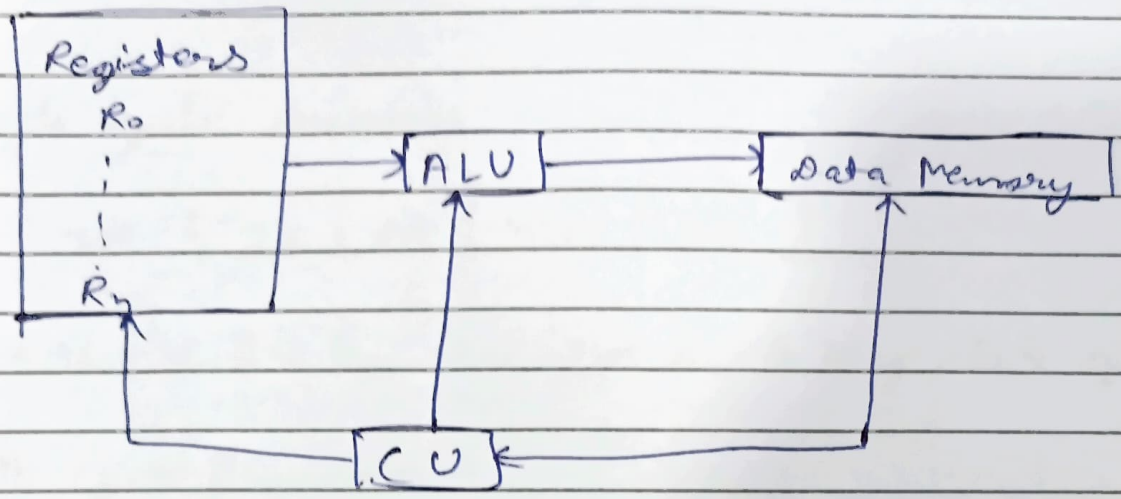
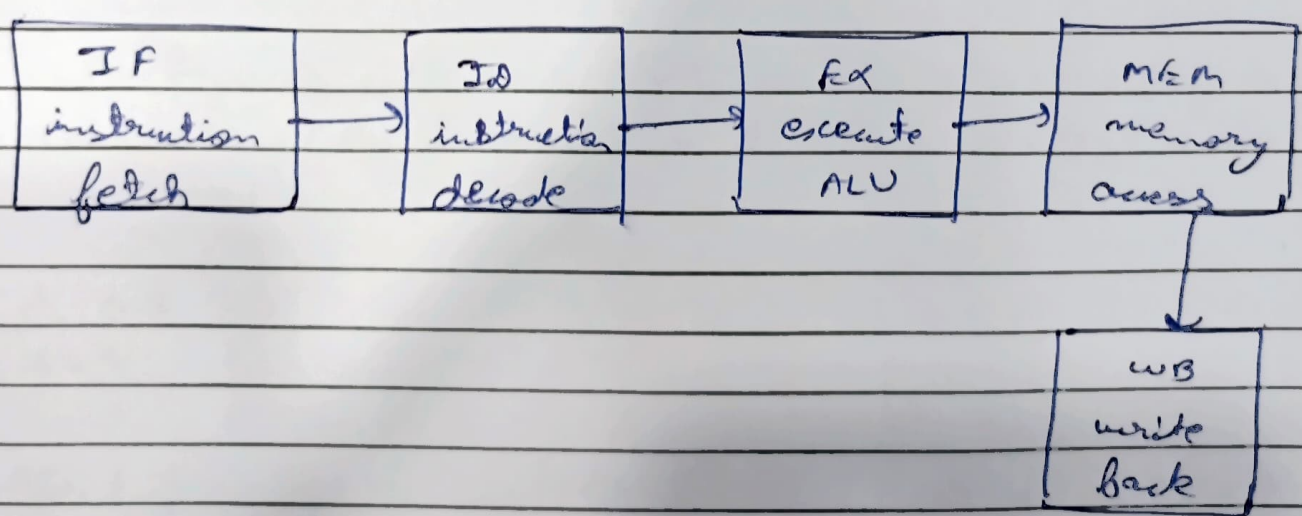
2)



Data flows from registers to ALU for processing, then to memory or back to registers. CU generates signals to control the whole operation.

3)



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4)

(a) Single cycle processor

IF	ID	EX	MEM	WB

entire instruction executed in 1 long clock cycle.

(b) Multi cycle processor

IF	ID	EX	MEM	WB

each stage is executed in different clock cycles.

SECTION - C

1) Data hazards will occur. They can be solved using data forwarding (by passing) or pipeline stalls (inserting bubbles).

2) $P = 0.8$
$S = 4$

$$\text{Speedup} = \frac{1}{(1-P) + (P/S)} \Rightarrow \frac{1}{0.2 + 0.2} \Rightarrow \frac{1}{5} \text{ or } 0.2$$

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SECTION-D

1)

IF → ID → EX → MEM → WB

EX stage: In this, ALU performs arithmetic or logical operations on the operands. It may also compute branch target addresses.
This stage is critical.

2) Example:

ADD R_1, R_2, R_3 ; $R_1 = R_2 + R_3$
SUB R_4, R_1, R_5 ; $R_4 = R_1 - R_5$

Prob: SUB needs R_1 before ADD writes it back.

Solⁿ: Use data forwarding from EX/MEM stage to EX stage or insert one stall

SECTION-E

1) Two assumptions while explaining pipelining:
- It increases instruction throughput by overlapping execution of multiple instⁿ.

2) Pipelining is the most challenging because it involves multiple stages, hazards & techniques to handle them, making it complex to visualize & implement.

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